



Service Tool Instruction

Service Tool Instruction
Number: 3377652-2

Revision Level:

Released Date: 01-aug-
2001

Description

Injection Timing Tool

Purpose

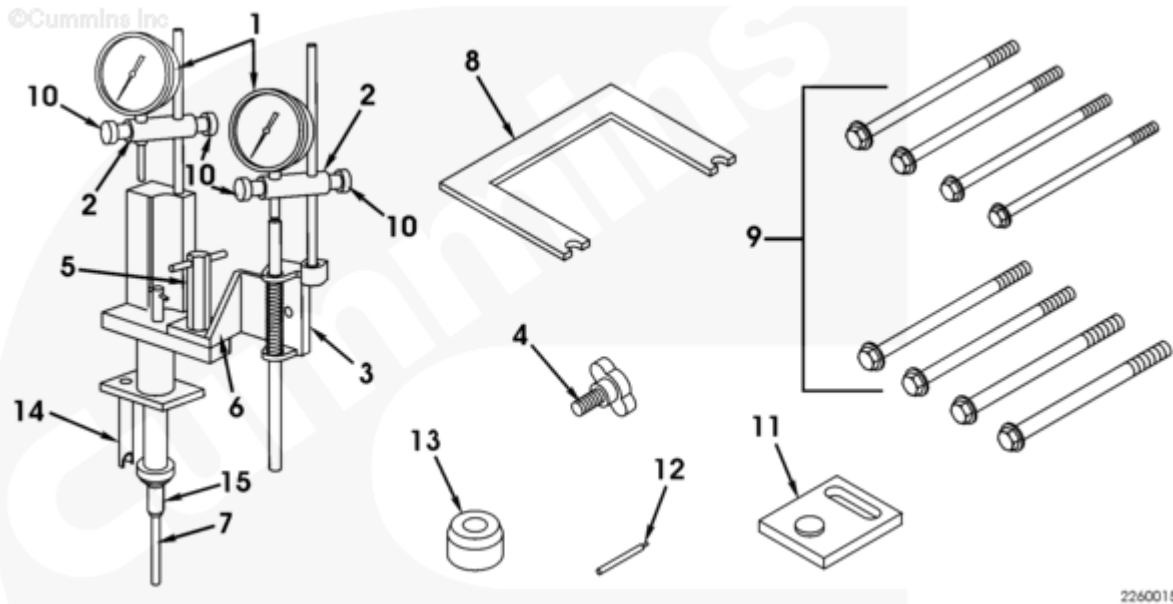
The injection timing tool, Part Number 3824942, is used to determine the relationship of the injector push tube travel to the piston travel so that fuel injection occurs at the proper time. It measures static injection timing on L10, M11, K19, QSK19, K38, and K50 engines.

The injection timing tool, Part Number 3823451, was superseded by Part Number 3824942, which contains additional parts to measure the QSK19 Series engines. If presently using the old tool, Part Number 3823451, order QSK19 update kit, Part Number 3824946, to update the present injection timing tool, Part Number 3824942.

Refer to Table 2 for items required for the QSK45, QSK60, and QSK78. These items are **not** included in the kit and **must** be purchased separately.

If presently using an injection timing tool with a locator spacer, Part Number 3824944, the new locator spacer, Part Number 3163304, **must** be ordered to replace it. The locator spacer, Part Number 3163304, drops into the injector bore and does **not** require the removal of the base tool collar.

The timing tool will operate with or without the rocker lever housing in position on the cylinder head. With the rocker lever housing in place, a slight misalignment of the push tube can occur because of the rocker lever position on some engines. This will **not** significantly affect the engine timing.



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Table 1, Injection Timing Tool, Part Number 3824942

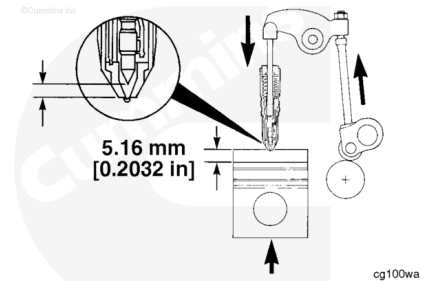
Item Number	Part Number	Description	Quantity
1	3376408	Dial indicator	2
2	3823517	Dial indicator support	2
3	3823518	Push tube plunger bracket	1
4	3376451	Push tube plunger knob	1
5	3376452	Locking nut	1
6	3376453	Center support bracket	1
7	3823519	Piston plunger bracket assembly	1
8	3824947	Distance-setting gauge, L10 and M11 engines	1
9	3823510	Hold-down capscrew, M8-1.25	1
9	3823511	Hold-down capscrew, 5/16-18	1
9	3823512	Hold-down capscrew, 3/8-16	1
9	3823513	Hold-down capscrew, 3/8-24	1
9	3823514	Hold-down capscrew, 1/2-13	1

9	3823515	Hold-down capscrew, 1/2-20	1
9	3823600	Hold-down capscrew, M10-1.25	1
9	3824943	Hold-down capscrew, 3/8-24 x 7-inch	1
10	3823530	Thumbscrew (4 shown)	6
11	3824945	Extension plate	1
12	3376409	Indicator stem extension	1
13	3163304	Locator spacer	1
14	-	Swivel clamp bracket	1
15	3823516	Timing fixture post locator	1
Not shown	3823268	Storage case	1
Not shown	3377652	Instruction sheet	1

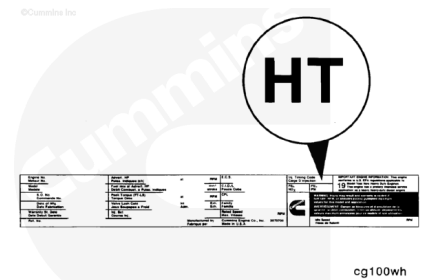
Table 2, Items Used with Injection Timing Tool, Part Number 3824942, Purchased Separately

Item Number	Part Number	Description	Quantity
Not shown	3163305	Indicator stem extension (QSK19, QSK45, QSK60, and QSK78)	1
Not shown	3163306	Distance-setting gauge (QSK19, QSK45, and QSK60)	1
Not shown	3163307	Hold-down capscrew, M10-1.25 x 165-mm (QSK45 and QSK60)	1
Not shown	3164220	Distance-setting gauge (QSK78)	1
Not shown	3164221	Hold-down capscrew, M10-1.5 x 170-mm (QSK78)	1

Injection timing refers to the amount of injection push tube travel (fuel injection stroke) remaining during the compression stroke when the piston is 5.16 mm [0.2032 in] before top dead center.



The engine timing code and nominal push rod travel are listed in the rear of the CPL bulletin. The two-letter timing code is found on the dataplate (as shown). This code then is used to determine the correct push rod travel.

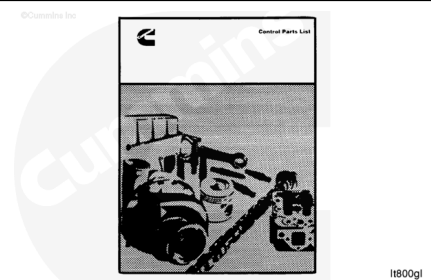


The timing **must** be checked on one cylinder for each camshaft follower housing on the NH, NT, and N14 engines, one cylinder on each bank of V-engines, and one cylinder on **all** other engines.

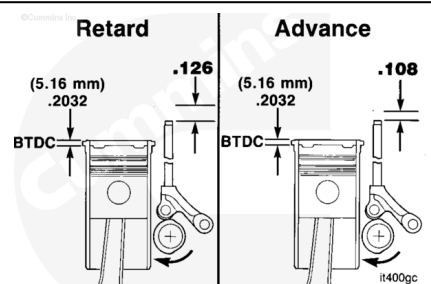
Injection timing can be adjusted by the following methods:

- **NH, NT, and N14 engines.** Change the camshaft keys (major) or camshaft follower box gasket thickness (minor). Increase gasket thickness to advance the timing; decrease the gasket thickness to retard the timing.
- **All other engines.** Change the camshaft keys.

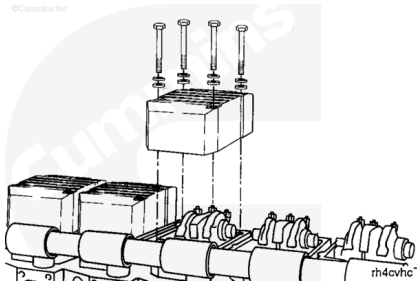
Refer to the appropriate engine shop manual, troubleshooting and repair manual, or Control Parts List, Bulletin 3379133, for available keys and gaskets.



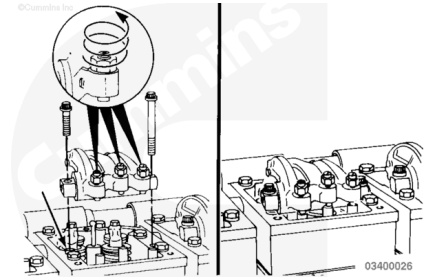
When timing values are high, the timing is retarded (push tube travel is insufficient, causing late injection). When timing values are low, the timing is advanced (push tube travel is excessive, causing early injection).



Remove the rocker lever or valve cover(s) from the cylinder head. Refer to the appropriate troubleshooting and repair manual or shop manual for the appropriate procedure.

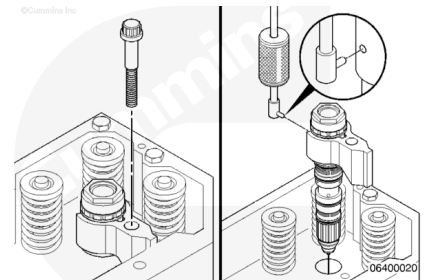


Remove the rocker lever assembly or assemblies. Refer to the appropriate troubleshooting and repair manual or shop manual for the appropriate procedure.



Remove the fuel injector(s). Refer to the appropriate troubleshooting and repair manual or shop manual for the appropriate procedure.

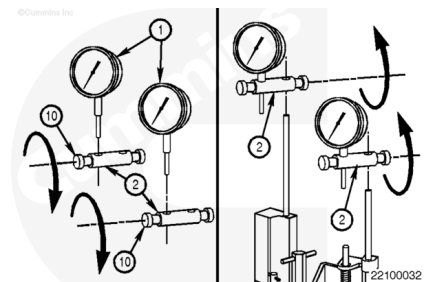
Note : It is not necessary to remove all of the injectors; however, engine rotation will be easier if all of the injectors are removed.



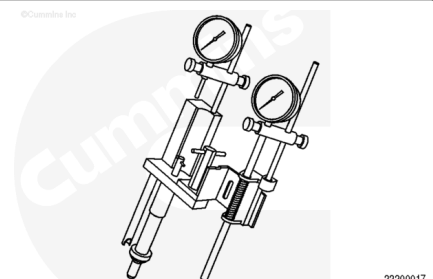
Install the dial indicators (1) through the dial indicator supports (2), and secure with the thumbscrews (10).

Install the indicator supports over the posts, and secure with thumbscrews (10).

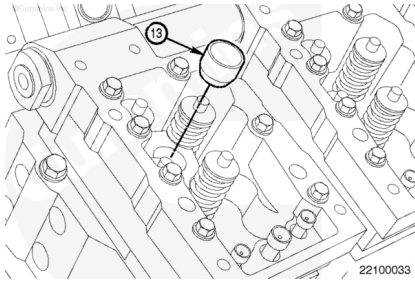
Pivot the stems away from the movable plunger posts. Add the appropriate indicator stem extension on the piston travel indicator, if needed.



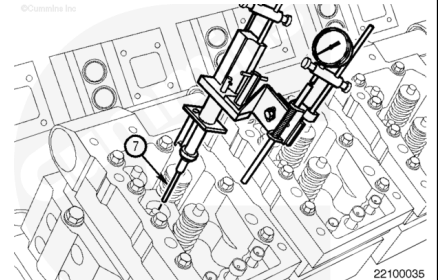
Attach the indicator stem extension, Part Number 3376409 or 3163305 to the piston plunger dial indicator. Indicator stem extension, Part Number 3163305, is used for the QSK19, QSK45, and QSK60 engines.



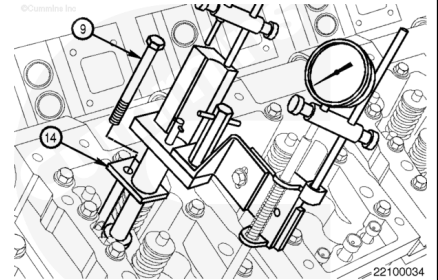
Insert the locator spacer (13) into the injector bore for the QSK19, QSK45, QSK60, and QSK78 Series engines.



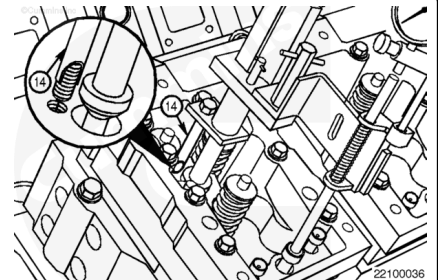
Place the piston plunger bracket assembly (7) into the injector bore.



Select the appropriate hold-down capscrew (9) that matches the injector clamp size and place into the swivel clamp bracket (14).



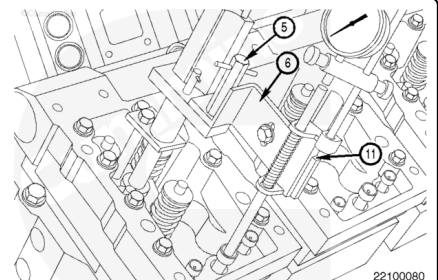
Align the swivel clamp bracket (14) with the injector clamp capscrew hole. Thread in the hold-down capscrew and finger-tighten it.



Some engines can require the use of extension plate, Part Number 3824945. Position the extension plate (11) between the bracket assembly and the push tube.

Position the push tube plunger post over the push tube. Adjust the center support bracket (6) spacing by loosening the locking nut (5). The push tube plunger assembly is positioned by visual alignment with the injector bore line of action.

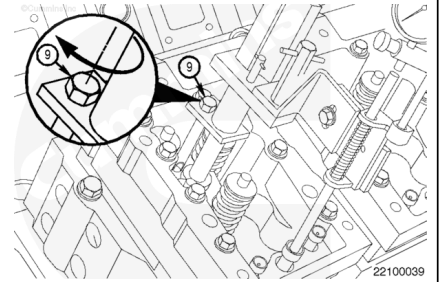
Finger-tighten the locking nut (5).



Tighten the hold-down capscrew (9). Do **not** overtighten the capscrew.

Torque Value: 11 n·m [97 in-lb]

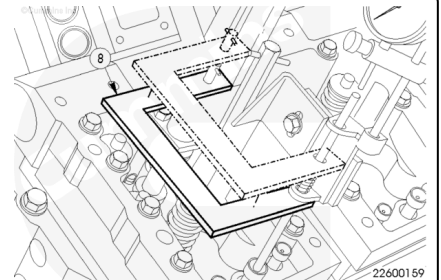
Make certain that the piston plunger post is rigid and clamped squarely to the head.



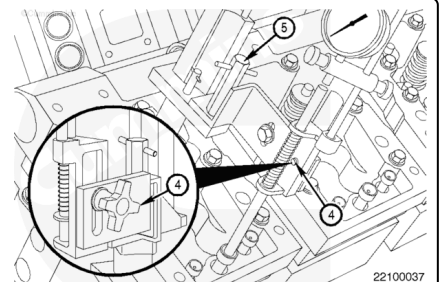
Use the appropriate distance-setting gauge (8) to set the distance between the piston and the push tube plunger posts.

To be certain of proper squareness, reposition the distance-setting gauge higher on the push rods. The injection rod is lifted to allow this check.

The plunger bracket is adjusted to allow spring compression of 13 mm [0.5 in]. Finger-tighten any connections that were loosened.



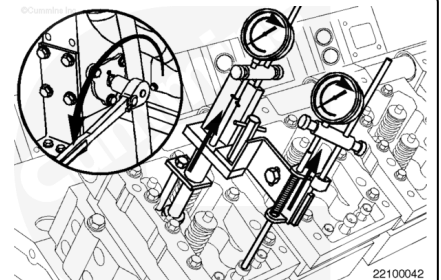
Once certain that the plungers are parallel, securely tighten **all** connections, including the push tube plunge knob (4) and locking nut (5).



Refer to the appropriate troubleshooting and repair manual or shop manual for the correct location to bar the engine.

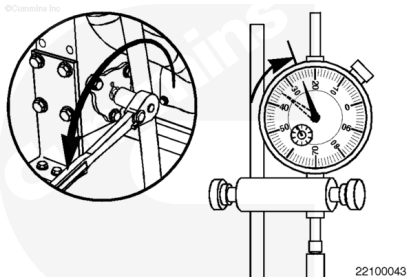
Bar the engine **clockwise** until both the timing and the push tube plungers move upward. This verifies the compression stroke. (Both indicators rotate in the same **clockwise** direction.) If **not**, rotate the crankshaft one more revolution.

Bar the engine **clockwise** until the piston plunger position is at top dead center of the compression stroke. Lower the dial indicator downward within 2.5 mm [0.1 in] of the fully compressed position.



Loosen the face thumbscrew of the dial indicator.

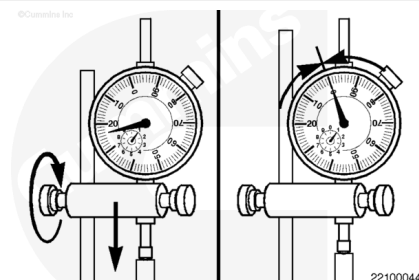
Rotate the dial indicator face until the dial indicator needle is pointing to the zero position.



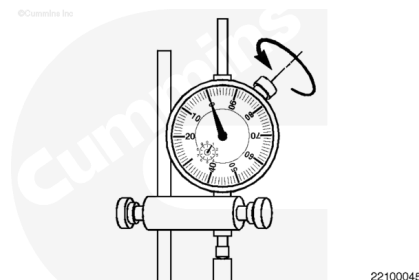
Bar the engine in a **counterclockwise** direction until the dial indicator needle moves approximately 1/4 turn in a **counterclockwise** direction (0.635 mm [0.025 in] before top dead center).

Bar the engine in a **clockwise** direction until the dial indicator needle moves **clockwise** and reaches top dead center; zero the dial indicator. Top dead center is indicated by maximum **clockwise** needle position of the piston plunger dial indicator.

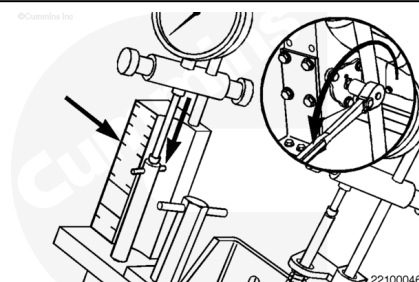
Zero the indicator **only** when the engine is barring in a **clockwise** direction.



Once top dead center has been determined, finger-tighten the face thumbscrew.



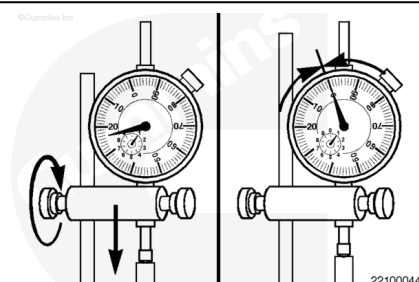
Bar the engine **clockwise**, or in the direction of engine rotation, until the piston plunger lowers to the mark on the timing tool that applies to the appropriate engine.



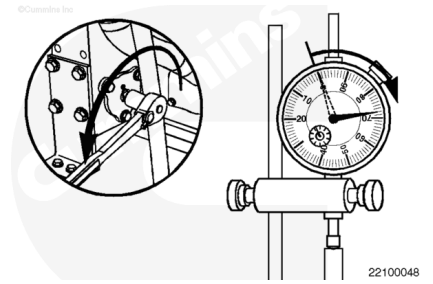
The distance traveled by the dial indicators is 2.5 mm [0.1 in].

Set the push tube dial indicator in line with the plunger post, and lower the dial indicator to within 2.5 mm [0.1 in] of the fully compressed position.

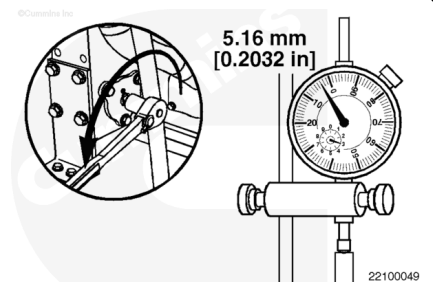
Zero the indicator by turning the dial indicator face to indicate the zero position.



Bar the engine **counterclockwise** to top dead center and continue **counterclockwise** through top dead center approximately 2-1/2 revolutions, 6.35 mm [0.250 in] on the piston gauge.



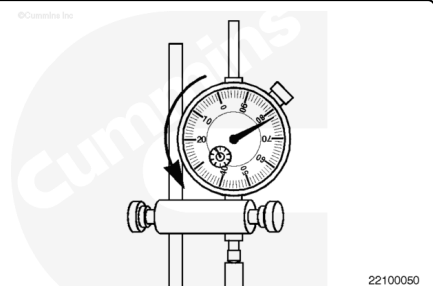
Rotate the engine in the direction of normal engine rotation until the piston dial indicator reading is 5.16 mm [0.2032 in] before top dead center.



Read the push tube dial indicator to determine the injector timing.

Tip

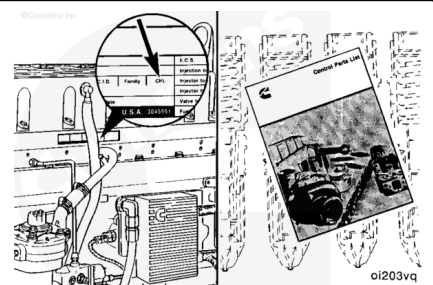
The dial indicator stem can be lifted back to the original position and lowered back down to verify the reading of the travel.



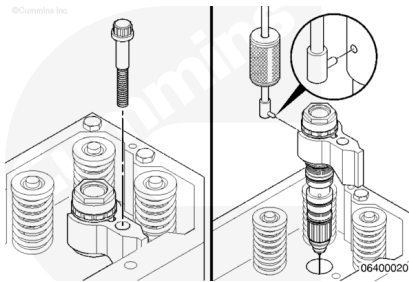
Use the engine dataplate and the appropriate engine CPL to compare the indicator reading with the engine's timing specification.

If the timing is **not** within engine specifications:

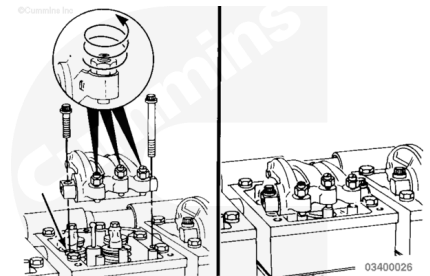
- Check the timing tool installation.
- Check the dial indicator readings.
- Check the crankshaft direction of rotation and timing sequence.
- Check the camshaft follower housing (NH, NT, and N14 engines) capscrew torque.
- Check the crankshaft and camshaft gear alignment.
- Repair the engine.



Install the injector(s). Refer to the appropriate troubleshooting and repair manual or shop manual for the correct procedures.



Install the rocker lever assembly or assemblies. Refer to the appropriate troubleshooting and repair manual or shop manual for correct procedures.



Install the rocker lever or valve cover(s). Refer to the appropriate troubleshooting and repair manual or shop manual for the correct procedures.

